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Original Article

Internet filters and entry pages do not protect children from online alcohol marketing

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Abstract We review programs and policies to prevent children from accessing alcohol marketing online. To update the literature, we present our recent studies that assess (i) in-built barriers to underage access to alcohol brand websites and (ii) commercial internet filters. Alcohol websites typically had poor filter systems for preventing entry of underage persons; only half of the sites required the user to provide a date of birth, and none had any means of preventing users from trying again. Even the most effective commercial internet filters allowed access to one-third of the sites we examined.

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Introduction

Alcohol and young people

Drinking among young people is a growing global public health problem. Australia's 2011 national school survey showed that 9.1 per cent of 12-year olds had consumed alcohol in the preceding month – 59.3 per cent of 17-year olds.¹ The National Alcohol Indicators Project estimated more than 80 per cent of alcohol consumed by 14–17-year olds was ingested in quantities posing short-term risk of injury. In Australia more than 3000 under-aged drinkers are hospitalized each year for alcohol attributable injury.² Regular alcohol consumption or binge drinking during adolescence predicts heavier alcohol consumption, binge drinking, and poor health outcomes later in life.^{3,4}

Alcohol advertising and young people

There is substantial and increasing evidence of an association between exposure to alcohol advertising and:

- Beliefs about the effects of alcohol (*expectancies*)^{5,6};
- Drinking intentions^{7,8}; and
- Current or future drinking.^{9,10}

Three longitudinal studies from the mid-2000s,^{6,11,12} now important for policy change, provided evidence of a direct relationship between advertising exposure and youth drinking.

Alcohol advertisements contain imagery and messages that Australian young people interpret to suggest that alcohol consumption will have positive psychological and social outcomes.^{13,14,15} New Zealand research has demonstrated that young people use alcohol (and alcohol brands) to communicate their identity; and that alcohol marketers are developing increasingly sophisticated messages and strategies to engage young people.^{16,17}

Young people on the internet

In 2010–2011, 93 per cent of Australian households with children under 15 had internet access, higher than childless households (74 per cent).¹⁸ In April 2009, 79 per cent of children aged 5–14 years accessed the internet²: 85 per cent were engaged in ‘educational activities’, but they also played online games (69 per cent), surfed the web (50 per cent), listened to or downloaded music (47 per cent), and used email (36 per cent).¹⁹

Alcohol marketing on the internet

The internet, a fast growing marketing medium, is increasingly used to advertise alcohol. Although it is difficult to determine exactly how much alcohol marketing appears on the internet,²⁰ in 2010, 27 of 40 (67.5 per cent) UK alcohol brands had a dedicated or shared (with other brands) website.²¹

Promotion of alcohol to underage people is a growing concern. A 2003 study of traffic to 55 alcohol company websites found many visits from underage people.²² These sites appeal to young people with



cartoons, interactive games, sport and music sections, downloadable content, social networking, and user-generated content.^{23–26}

Although alcohol marketing on the internet takes many forms, including alcohol-related pages on social networking sites, our study focused on brand websites, as:

- they are clearly defined as ‘marketing communications’, within the control of the alcohol company, and subject to regulations on alcohol marketing;
- age restrictions on access are entirely within the control of the brand, rather than the platform;
- companies consider them to be of primary importance; and,
- if we are to address social networking, we need to effectively control exposure of young people to these ‘old school’ marketing communications.

Companies themselves control format and content of alcohol brand websites including restrictions on underage access. Social media platforms, such as Facebook and YouTube, require users to create a profile and provide personal information, including date of birth.^{27,28} Facebook permits no one under 13 years of age to create a profile; its explicit guidelines ensure that alcohol advertisements, sponsored stories, and alcohol brand pages are closed to users whose profile places them under the legal age for alcohol purchases (18 years in Australia).²⁹ Facebook does not verify age; thus someone under 18 could deliberately create a false profile to access an alcohol brand page, rather than come across an alcohol brand website ‘accidentally’ or via a search engine.

A 2013 survey of 200 US marketers in six industries concluded that ‘the corporate website will not be replaced by a brand’s social media presence anytime soon’.³⁰ A UK cross-sectional survey of 12–14-year olds found current drinkers ($n = 318$) were more likely to be aware of alcohol social networking sites than brand websites (22 per cent compared with 7 per cent); but this was not the case for current non-drinkers ($n = 594$) (7 per cent compared with 6 per cent).³¹ An Australian study with 12–17-year olds found that 55 per cent had seen alcohol advertising on the internet.³²

Interactivity on social networking sites encourages alcohol marketing. Consumers can co-create marketing messages and act as brand ambassadors. These phenomena should be studied, but platforms change

rapidly. A 2013 report by Pew Research found young people moving away from Facebook towards Twitter and Instagram as older people (including their parents) join Facebook.³³ Understanding alcohol brand websites, plus the effectiveness of filters and age restrictions, can focus research and aid policy development for alcohol marketing on social networking platforms.

Content control, the role of filters, and age verification pages

Internet data are growing beyond the capacity of individuals, organizations, or government to understand or monitor content. WorldWideWeb-Size.com estimated the web contained 8.1 billion pages on 19 November 2012; the Next Web group, 48.4 billion to 121 billion indexed web pages.^{34,35} Time magazine's *Techland* website asserted that 'Every hour, enough information is consumed by internet traffic to fill 7 million DVDs. And four years from now, it'll be four times larger than that'.³⁶

In the 1990s, people began looking for ways to protect themselves and children from inappropriate content. They designed internet filters to enable parents (or, in some cases, governments, or other agencies) to block specific content. A user must nominate explicit terms or categories to block – such as 'alcohol products', 'alcohol drinking', 'drugs' or 'drug usage'. A filter can also be set up to block entire *domains*, such as YouTube or Facebook, and/or internet *features* such as chat or instant messaging. Filters assure that users will be unable to access particular content that a parent deems inappropriate for her/his child. Transparency Market Research (a United States-based market intelligence company) valued the content filter market at US\$1.2 billion in 2008, and projected that it will increase to \$6.7 billion by 2018.³⁷

Alcohol advertisers in Australia claim that an *age verification page* (AVP) can keep young people from alcohol websites by requiring that they confirm they are of a certain age before they can enter the website, but no requirement exists in Australia, or elsewhere, to ensure that AVPs are effective. Such mechanisms are probably inadequate; underage people can easily circumvent this barrier.³⁶

Purpose of the studies

Exposure to alcohol marketing (including online) affects youth drinking behaviors.^{21,31} Thus burgeoning internet alcohol marketing carries



serious implications for public health. No one knows what proportion of young people have accessed alcohol advertising websites, but the fact that most teenagers (and over half of 5- to 8-year olds) go online means that we need to examine strategies to prevent or reduce access.

We report on two studies in Australia that examined the existence and effectiveness of tools to prevent children from accessing alcohol marketing online. In Study 1, we examined strategies used by alcohol marketers to prevent those under legal drinking age from accessing their product websites; in Study 2, we examined the effectiveness of commercial filters in restricting access to online alcohol marketing.

Methods

Study 1: Effectiveness of alcohol brand websites' own filters/entry controls

Based on market trends in Australia,^{38–41} we created a sample of 25 alcohol brand websites. We used population-level sales, as information by market segment was not publicly available. The 2011 national school survey (ASSAD) found that 37 per cent of 12–17-year olds usually drank pre-mixed spirits (*alcopops*), 27 per cent other spirits, and 18 per cent beer. We selected alcopop brands using data from an audit of bottle shops in regional (large cities/towns that are not major cities; based on the Australian Standard Geographical Classification) and urban areas of New South Wales, Australia.⁴² Our final list of brand websites appears in Table 1. Reflecting drink preferences of young people, it included nine beer, three alcopops, and 13 spirit brands (10 of which also produce a pre-mixed version).

Between 1 and 8 March 2011, trained coders described the websites, recording whether there was an *entry page* requiring age; its characteristics; and the site's response to entry attempts by a (pseudo) underage consumer.

Study 2: The effectiveness of commercial internet filters

We selected five commonly used internet filters from a list of 'family friendly filters' provided by the Australian Internet Industry Association and reviews of internet filters provided by the US-based 'Top Ten Reviews'.^{43,44} Our final choice of filters was based on costs, availability

Table 1: Alcohol brands coded in Study[†]

<i>Product name</i>	<i>URL</i>	<i>Country of site</i>
Bacardi Rum	www.bacardi.com	UK
Baileys Irish Cream	www.baileys.com	Australian version of international site
Bundaberg Rum	www.bundaberg.com.au	Australia
Carlton Draught	www.carlton.com.au	Australia
Coopers Ale	www.coopers.com.au	Australia
Corona Extra	www.corona.com	US/international
Cougar Bourbon	www.fostersgroup.com/brands/	Australia
Cruiser	www.cruiserrange.com.au	Australia
Hahn Superdry	www.hahnsuperdry.com.au	Australia
Jack Daniels	www.jackdaniels.com	US
Jim Beam Bourbon	www.jimbeam.com	US
Johnnie Walker Scotch	www.johnnie-walker.com	Australian version of international site
Jose Cuervo Tequila	www.cuervo.com	US
Midori	www.adoremidori.com	Australia
Pulse	www.independentdistillers.com	Australia
Pure Blonde	http://cub.com.au/beer/	Australia
Ruski	www.vok.com.au/ruski-premix.html	Australia
Smirnoff Vodka	www.smirnoff.com	Unknown location (Verisign)
Southern Comfort	www.southerncomfort.com	Australian version of international site
Tooheys Extra Dry	www.tooheys.com.au	Australia
Toohey New	www.tooheysnew.com.au	Australia
Victoria Bitter	www.vb.com.au	Australia
Wild Turkey	www.wildturkey.com.au	Australia
Woodstock Bourbon [†]	www.independentdistillers.com	Australia
XXXX	www.xxxx.com.au	Australia

[†]Note that while not available at the time of our data collection, at the time of writing this brand now has a website (<http://www.woodstockbourbon.com.au/babes.html>) that consists exclusively of photos and video of the ‘Woodstock Babes’ (semi-clad women posing suggestively for a calendar) and a link to the Woodstock Facebook page. This website has no age restrictions/entry page.

to Australian consumers, and our ability to install and run them within our computing environment. We tested Net Nanny 6.5, Pure Sight PC, Content Keeper Home Edition, McAfee SmartFilter, and Trend Micro Titanium Internet Security 2011.

We used six identical Dell desktop PCs running Microsoft’s Windows XP operating system and Internet Explorer 8 web browser, installing one filter on five machines. The sixth (control condition) had no filter. One of us [JT] set up the internet filters using standard default settings. Where



configuration was required, he selected the 'teen' setting. Where two settings covered 11–17-years olds, he selected the younger age bracket. All had a content category 'Addictions' with specific classifications for 'Drugs' and 'Alcohol and Tobacco'. Where a filter could be disabled or bypassed, he enabled 'Always On'. On the first test session, we updated each online to their latest version.

Participants

We tested effectiveness twice for each filter and for the control – once by a research center staff member (an adult with a graduate degree in a health discipline), and once by an undergraduate student from the Informatics Faculty. We used undergraduates because they were close in age to the target group, had similar levels of computer skills – more advanced skills than most adults; and, like teenagers, were motivated to try to 'beat' the filters. Six staff members and six students participated.

It took approximately 45 min to test each participant. We gave them each a AU\$20 department store gift voucher. Testing took place from 2 to 13 May 2011, at a university computer laboratory.

Materials

We developed a two-part questionnaire:

Part A to learn through responses to open-ended questions how participants would go about finding alcohol brands on the internet. We undertook searches using those terms, and noted the URLs for the first five websites generated.

Part B contained binary response and Likert-style items for the 25 alcohol brands in Study 1. We instructed participants to try to access the websites for each of the 25 alcohol brands and answer questions about their ability to do so. One project staff member had pilot tested the instrument and found it to generate usable data for our analysis.

Results

Study 1: The effectiveness of alcohol brand website own filters/entry controls

Alcohol websites typically had poor filter systems for preventing entry of underage persons. Most had an entry page (22 of 25), but three sites were accessible with no entry page requirements. Seven of 22 sites did

not request a full date of birth; they merely provided a statement (for example, 'You must be aged 18 years or over to view this site'), with a button click response of 'Yes, I am' or 'No, I'm not'. Detailed age filters (asking for a full date of birth and then denying entry if the age entered was less than 18 years) were present in only 13 of the 25 sites. Two sites requested a full date of birth, but allowed access regardless of the age entered. None of the sites prevented users from trying again with a different date of birth once access had been denied. Some allowed users to return straight back to the login page by clicking a link or by clicking 'back'.

Study 2: The effectiveness of commercial internet filters

Finding alcohol websites generally

Part A tested internet filters' capacity to block internet searches using alcohol-related words generated spontaneously by participants. All 12 participants told us that to find alcohol brands on the web, they would type search terms, such as the brand name, into the Google's search engine, describing this as the most common way to find this information.

Only Net Nanny adequately blocked these searches. With Net Nanny active, search terms such as 'alcohol', 'beer' and 'spirits' generated no results but 'drinking games' offered access to 'idrink.com/games' and '50 drinking games guaranteed to get you hammered: www.amog.com/lifestyle/'. Thus while Net Nanny blocked the most obvious search terms, less obvious combinations allowed users to find websites promoting dangerous drinking practices.

Finding and accessing 25 specific alcohol brand websites

We found variable effectiveness preventing access to specific branded sites (see Figure 1). Net Nanny and McAfee were most effective preventing users from *finding* alcohol websites. Pure Sight and Content Keeper appeared to be ineffective in preventing users from finding alcohol brand websites. Trend Micro gave mixed results, perhaps reflecting the computer skills of the two testers. With Net Nanny active, the staff member found four of the 25 sites, the student 11; with McAfee, the staff member found eight sites and the student five. As expected, the control condition allowed unrestricted access to alcohol brand websites.

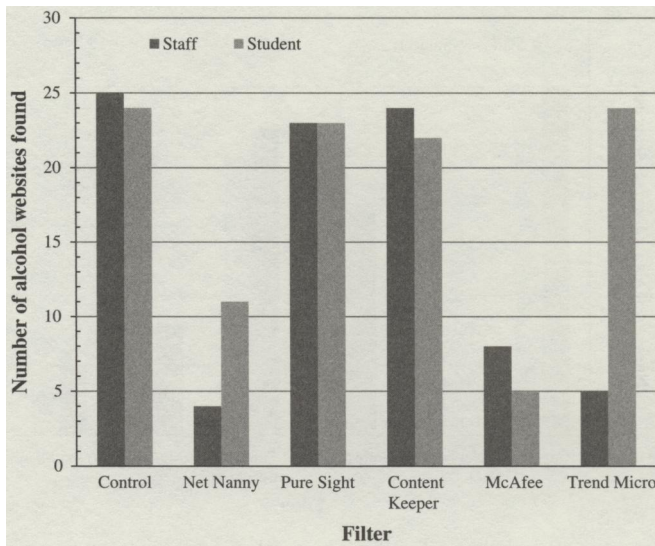


Figure 1: The number of alcohol websites found when using the internet filters compared with the control condition.

On finding alcohol brand websites, participants attempted to enter through the ‘login’ or ‘entry’ page. In general, the internet filters were slightly more effective at preventing actual entry to the main site (see Figure 2).

We also measured how quickly, with the filter active, testers could access a site from when they entered the brand name into the search engine. As expected, the control condition provided the easiest and quickest access to sites.

Discussion

The tokenistic nature of brand sites’ entry controls

Age verification requirements for online services and websites are vexing. The UK’s Portman Group advocates that AVPs require the user to ‘actively input their date of birth’ as this is a more ‘stringent’ approach than the user simply selecting an option that they are over 18 (Portman Group pp18).⁴⁵ More active approaches – such as drawing on publicly available databases to confirm identities – have been shown to be more effective but raise privacy issues. They may conflict with efforts to

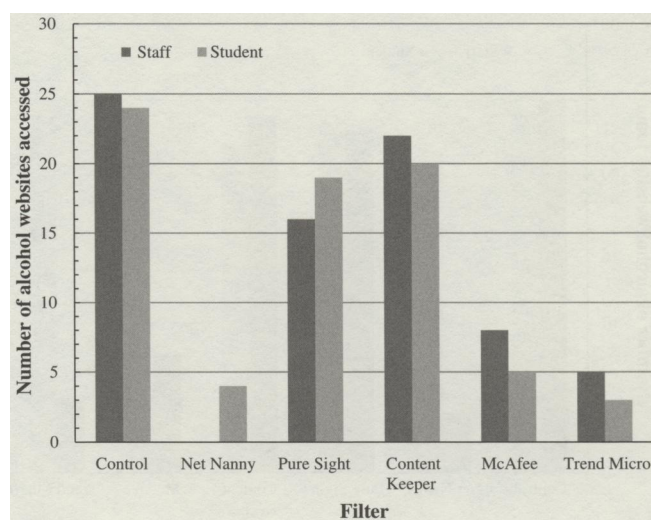


Figure 2: The number of alcohol brand websites accessed when using the internet filters compared with the control condition.

prevent the alcohol industry from collecting personal data from consumers, particularly underage ones. The United States Federal Trade Commission's (FTC) Children Online Privacy Protection Act (COPPA) 1998, for example, places stringent privacy and security measures around requesting data from children. The FTC suggests that firms use syndicated or 'independent demographic surveys' to ensure they do not place promotions in online locations accessible by US computers with a teen user base greater than 30 per cent, and where this is not possible that they implement a third-party age verification system.⁴⁶

Australia's Alcohol Beverages Advertising Code (ABAC), which covers internet advertising, is silent on age verification.⁴⁷ A 2005 survey of Australian adults found that 82 per cent thought age verification insufficient to prevent those aged under 18 from entering an alcohol brand website. Our findings support this perception.⁴⁸ The 25 alcohol websites had poor filter systems; none had any means to prevent users from trying again once access had been denied.

The limited effectiveness of commercial filters

We tested five commonly recommended internet filters and despite claims to enable parents to prevent their children accessing undesirable



websites, several appeared ineffective in blocking access to alcohol brand websites. All were more effective than the notional efforts made by the sites themselves.

A similar study of US alcohol brand sites, conducted by the Center on Alcohol Marketing and Youth (CAMY) in 2003, found similarly wide variation in the effectiveness of filters.²³ The two commercial filters we found most effective, Net Nanny and McAfee, had also been tested 8 years earlier by CAMY, but found to be the least effective (blocking 6 and 0 per cent of the sites respectively). Perhaps there have been substantial improvements to these filters between the two studies.

One of the filters we examined (Net Nanny) had the additional feature of regular notification to the 'parent' (in this case, the staff member who set up the filters) when their 'child' (the tester) visited inappropriate sites, suggesting that this filter could be used in conjunction with parental discussions with children about their internet use. The filter notifies both access to known sites containing alcohol content and other references to alcohol as defined in the filter settings. Notification may be important as filters were effective in blocking searches using specific words, but allowed access to other websites, such as those promoting drinking games.

Changes in internet usage

Our study used technology similar to current configurations in schools and many households as measured in the 2009 Australian Census.¹⁹ With a change in the way the internet is used and accessed, filters on fixed desktop computers are becoming obsolete. Movement toward mobile technologies – smartphones, tablets, and laptop computers – allows users to access information from anywhere, at anytime, and often through dedicated applications like Facebook, YouTube, and Four-Square. Facebook had one billion registered users as of October 2012, with 604 million accessing the site from a mobile device during that time.⁴⁹ Facebook is integrated with over one million websites, contains three million fan-pages and 900 million objects for users to interact with, including photos, videos, and posts. As a network it represents over 100 billion 'friendship links' that share over one billion pieces of content daily. In 2012, 80 per cent of businesses used Facebook and 39 per cent planned to spend money marketing on the platform.⁵⁰

In Australia, Facebook has 11.7 million users – approximately 55.2 per cent of the population.⁵¹ Socialbakers, a social media analytics company with more than 1000 clients in more than 75 countries, ranks Smirnoff Australia, Bundy Rum, Baileys Australia, Pure Blonde, Yellow Tails, and Jacobs Creek in their Top 100 brands to be ‘recommended’ among Facebook ‘friends’ in Australia.⁵² The ability to connect from any location, given that 40 per cent of children aged 6–10 use a social media site and 31 per cent a chat feature, means that Facebook provides massive potential for access to inappropriate content, and renders ineffective even the most effective of the filters we tested.⁵³

We tested filters designed to restrict websites accessed through a browser using a desktop PC as used in schools and households at the time. Products specific to new applications (such as Facebook) and devices (such as smartphones) exist. Growth in the internet filter market follows increasing use of mobile technologies and new demand for content safe environments. New filters will constantly need to be developed for new platforms and new devices, making regulation of the content itself even more important if we are to reduce young people’s exposure to alcohol marketing.

Limitations

The results should be seen as indicative rather than definitive. Only two individuals tested each of the filters, one staff member and one undergraduate student, and possibly the level of computer skills caused variation between filters. Our tests probably over-estimate the effectiveness of the filters in blocking deliberate access to alcohol websites. They may be effective in blocking accidental access to websites promoting alcohol, but may not withstand the concerted efforts of a teenager with time and computer skills to search many terms.

Conclusion

Many have questioned the effectiveness for traditional media’s self-regulatory system for alcohol advertising in Australia.^{13–15,54,55} However flawed these systems, they do provide some limits on the nature and amount of alcohol advertising to which young people are exposed. Our study shows that absent controls on alcohol advertising on



the internet, a vast amount of alcohol marketing remains available to audiences of all ages at all times.

Young people's use of and comfort with the internet increases rapidly and internet filters have been proposed as a way to prevent children from accessing potentially harmful content, such as alcohol advertising. Our research makes understanding of the effectiveness of filters more robust by demonstrating that both industry filters (landing pages) and commercial filters (software installed on computers to prevent access to specific content) are largely ineffective in preventing children and adolescents from accessing alcohol websites. Given the direct association between exposure to alcohol advertising and adolescent alcohol consumption, new policies are needed to regulate alcohol advertising on the internet.

A more measured approach would require more effective filters for end-users (parents in particular), recognizing that the effectiveness of external regulation of the online environment relies on consumers and parents acting to protect themselves and their children plus commitment from industry to develop and implement better age-based entry restrictions.

A regulatory approach combining government oversight and enforcement of industry-led practice standards is needed. The standards should prohibit industry from using techniques and approaches that appeal to children and young people, not just standards to prevent exposure through effective age filters. Without a multi-strategy approach, we will see a continuing increase in rates of alcohol consumption among our young people.

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